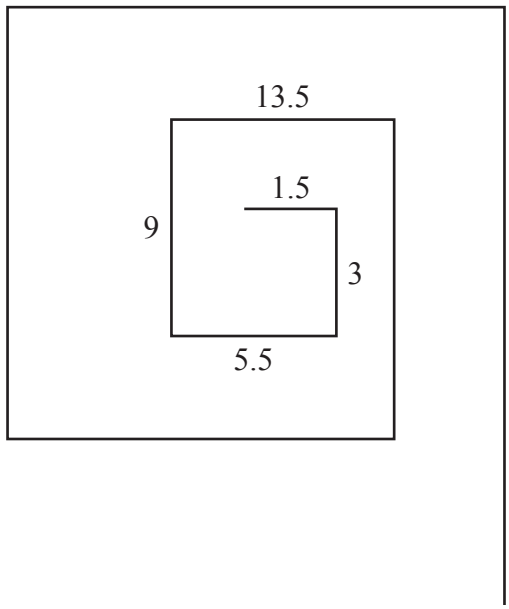
18 The diagram shows the first 10 sides of a spiral pattern.
It also gives the lengths, in cm, of the first 5 sides.



The lengths, in cm, of the sides of the spiral form a sequence.

Find an expression in terms of n for the length, in cm, of the nth side.

n	1	2	3	4	5
Length	1.5	3	5.5	9	13.5
1st diff.	1.5	2.5	3.5	4.5	
2nd diff.	1	1	1		

1 mark
Common second difference = 1

Coefficient of $n^2 = 1 \div 2 = \frac{1}{2}$.

1 mark
$\frac{1}{2}n^2$ term

$\frac{1}{2}n^2$ gives 0.5, 2, 4.5, 8, 12.5: add 1 each time.

nth side = $\frac{1}{2}n^2 + 1$ cm.

1 mark
$\frac{1}{2}n^2 + 1$

(Total for Question 18 is 3 marks)